CHAPTER 26

Library Reference: Core Data Types

In this chapter, we'll take a closer look at Ruby's core data types. The goal is to give you more information about what you can do with these classes, and also to discuss related functions together so that you can browse and perhaps find a new feature that might help. We're presenting the topics in alphabetical order for easier browsing, and we'll cross-reference between topics as needed.

This is not intended to be a complete listing of every class, every method, or every option. For that, please refer to the official Ruby documentation at https://docs.ruby-lang.org.

Dates and Times

Ruby has three separate classes to represent date and time data: Time, Date, and DateTime.

- Time represents a specific moment in time, and you can retrieve both date and time information based on that. The Time class is based on a library that is common to Unix systems and is used by many programming languages.
- Date represents a date only with no time information attached. It is useful for calendar arithmetic that does not depend on time of day.
- DateTime also represents a specific moment in time but uses a different internal representation than Time. DateTime is now considered deprecated—at one point it had a more complete API than Time, but that is no longer true. Currently the only recommended use of DateTime is if you are dealing with dates in the distant past.

Creating Time Instances

There are a few ways to create a new Time instance. The method you are most likely to see is Time.new, with no arguments, which is also aliased as Time.now, and which returns the current time. For example, to see when this section was added to the book:

```
Time.now  # => 2023-02-26 15:56:02.299719 -0600
```

By default, the resulting time is in the current time zone. To change the time zone, you use an optional keyword argument, in, that takes the zone, as in Time.now(in: "-05:00").

You will see a few time methods that allow you to specify a time zone. For those methods, Ruby allows a time zone to be specified by: